

LangChain & GenAI Engineer Curriculum (Headings and Subtopics Only)

1. Generative AI Foundations

- LLMs Overview
- Transformers
- Tokenization
- Embeddings
- Prompt Engineering
- Inference Parameters
- Context Windows
- Model Evaluation
- Open-source vs Proprietary Models

2. Python for GenAI

- Python Fundamentals
- OOP
- Async Programming
- Virtual Environments
- Package Management
- API Integration

3. LangChain Fundamentals

- Architecture Overview
- Models
- Prompts
- Messages
- Chains
- Runnable
- Output Parsers
- Callbacks

4. Prompt Engineering with LangChain

- PromptTemplate
- ChatPromptTemplate
- Few-shot Prompting
- Example Selectors

- Prompt Composition

5. Models & Providers

- OpenAI Integration
- Anthropic Integration
- Google Gemini Integration
- Local Models
- Model Configuration
- Streaming Responses

6. Output Handling

- String Output Parsing
- Structured Output
- Pydantic Output Parsers
- JSON Mode
- Function Calling
- Tool Calling

7. Retrieval-Augmented Generation (RAG)

- RAG Fundamentals
- Document Loading
- Text Splitting
- Chunking Strategies
- Embedding Models
- Vector Databases
- Retrievers
- Hybrid Search
- Reranking

8. Document Loaders

- PDF Loaders
- CSV Loaders
- Word Documents
- HTML/Web Loaders
- JSON Data
- Custom Loaders

9. Text Splitters

- Character Splitters
- Recursive Splitters

- Token-based Splitting
- Semantic Chunking

10. Embeddings

- Embedding Theory
- OpenAI Embeddings
- Sentence Transformers
- Embedding Evaluation

11. Vector Databases

- FAISS
- Chroma
- Pinecone
- Weaviate
- Qdrant
- Milvus

12. Retrievers

- Vector Store Retrievers
- Multi-Query Retrieval
- Parent Document Retrieval
- Contextual Compression
- Ensemble Retrieval

13. Agents

- Agent Fundamentals
- Tool Usage
- ReAct Pattern
- Agent Executors
- Multi-tool Agents
- Planning Agents

14. Tools Integration

- Custom Tools
- Web Search Tools
- Database Tools
- API Tools
- Calculator Tools

15. Memory Systems

- Conversation Memory
- Buffer Memory
- Summary Memory
- Persistent Memory

16. LangGraph

- Graph Fundamentals
- Nodes & Edges
- State Management
- Conditional Routing
- Human-in-the-loop
- Multi-Agent Systems

17. Multi-Agent Architectures

- Agent Collaboration
- Supervisor Agents
- Specialized Agents
- Agent Communication

18. Structured Data & Databases

- SQL Databases
- SQL Agents
- PostgreSQL
- SQLite
- NoSQL Integration

19. APIs & MCP

- REST APIs
- FastAPI
- Model Context Protocol (MCP)
- Tool Servers
- Client-Server Architecture

20. Evaluation & Observability

- LLM Evaluation
- RAG Evaluation
- Tracing
- Monitoring
- LangSmith
- Error Analysis

21. Deployment

- Docker
- Cloud Deployment
- Serverless Deployment
- Scaling
- CI/CD Pipelines

22. Production GenAI

- Caching
- Rate Limiting
- Cost Optimization
- Latency Optimization
- Guardrails
- Reliability

23. AI Safety & Security

- Prompt Injection
- Data Privacy
- Hallucination Mitigation
- Content Filtering
- Access Control

24. Advanced RAG

- Graph RAG
- Agentic RAG
- Multimodal RAG
- Knowledge Graphs

25. Multimodal AI

- Image Understanding
- Vision Models
- Audio Processing
- Video Processing

26. Fine-Tuning & Custom Models

- Fine-Tuning Concepts
- LoRA
- QLoRA
- Model Adaptation

27. MLOps for GenAI

- Experiment Tracking
- Model Versioning
- Monitoring
- Governance

28. Capstone Projects

- Chatbot
- Enterprise RAG System
- Agentic Assistant
- Multi-Agent Workflow
- Document Intelligence Platform